# UNIT 2 – Data Storage Technologies

## Activity 2: Large-Scale E-Commerce Sales and Customer Behaviour Analysis Using Hadoop MapReduce

**CO Mapped:** CO3 – Develop a MapReduce program
Type: Practical + Analytical Activity

## Objective

Develop a Hadoop MapReduce application using Python Hadoop Streaming to analyse large-scale e-commerce transaction data, summarize customer purchasing behaviour, calculate spending metrics, classify customers, and use a Combiner for local aggregation to reduce shuffle data. MapReduce program for e-commerce transaction data analysis using Hadoop. Experiment with performance tuning techniques and document findings with visualizations.

## Tasks to Perform

### Task 1: Customer-wise Transaction Analysis

- For every customer, calculate total number of transactions.
- Calculate total purchase amount across all transactions (Quantity Purchased × Unit Price).
- Calculate number of successful transactions.
- Calculate number of failed transactions.

### Task 2: Customer Spending Analysis

- For every customer, count high-value transactions with transaction amount ≥ ₹50,000.
- Calculate total amount spent on successful transactions only.

## Task 3: Customer Classification

- Classify as PREMIUM when total successful purchase amount ≥ ₹1,00,000.
- Classify as FREQUENT when successful transaction count is at least 4.
- Classify as NORMAL when neither condition is satisfied.
- Classify customers meeting both conditions as PREMIUM_FREQUENT.

## Task 4: Combiner Implementation and Performance Requirement

- Implement a Combiner to locally aggregate transaction statistics before intermediate data is transferred to Reducers.
- Ensure the Combiner is safe for repeated aggregation and explain how it reduces shuffle data.

## Task 5: Hadoop Streaming Execution and Validation

- Develop Python mapper, reducer, and Combiner scripts and configure Hadoop Streaming.
- Execute the application using HDFS input/output paths on the attached sample dataset.
- Validate results for selected customers and handle malformed records appropriately.

## Task 6: Results Analysis and Documentation

- Present customer-wise metrics and classifications in clear structured output.
- Compare execution with and without Combiner where feasible; record runtime and shuffle observations.
- Document methodology, sample results, validation, commands, and key observations in a concise report.

## Dataset

- The sample dataset is attached with this activity. Use it for development, testing, and validation. Fields: Transaction ID, Customer ID, Product Category, Quantity Purchased, Unit Price, Transaction Status, Payment Mode.

## Deliverables

Students submit the following items:

- Python Hadoop Streaming mapper, reducer, and Combiner source code.
- Hadoop Streaming execution commands/configuration and HDFS input/output evidence.
- Customer analysis and classification output with validation evidence.
- Performance comparison with/without Combiner (where feasible).
- Concise report containing approach, sample results, and observations.
- Link to the GitHub repository containing code ,data used, and report.

# RUBRIC – E-Commerce Sales and Customer Behaviour Analysis (Activity 2)

| Task / Criterion | Excellent (4 Marks)<br>A student will achieve this level when… | Good (3 Marks)<br>A student will achieve this level when… | Satisfactory (2 Marks)<br>A student will achieve this level when… | Poor (1 Mark)<br>A student will achieve this level when… | No Attempt (0) |
|---|---|---|---|---|---|
| Task 1: Customer-wise Transaction Analysis | Correctly computes all four customer-level metrics for every customer; records and statuses handled accurately. | Computes all metrics correctly with minor formatting or edge-case issues. | Computes most metrics, but one metric or some customer results contain errors. | Output is incomplete or has major aggregation errors. | No customer-wise analysis submitted. |
| Task 2: Customer Spending Analysis | Accurately counts transactions ≥ ₹50,000 and totals successful purchase amounts per customer. | Both calculations are mostly correct with minor errors. | Completes one calculation correctly or has several inaccuracies. | Calculations are largely incorrect or incomplete. | No spending analysis submitted. |
| Task 3: Customer Classification | Applies all conditions correctly, including PREMIUM_FREQUENT and NORMAL classification. | Classification is correct with minor edge-case issues. | Applies basic conditions but misses a rule or combination. | Classification logic is substantially incorrect. | No classification submitted. |
| Task 4: Combiner Implementation | Implements safe local aggregation and demonstrates/explains reduction of intermediate shuffle data. | Combiner works correctly with a reasonable explanation. | Combiner is present but aggregation or explanation is limited. | Combiner is incorrect, unsafe, or not meaningfully used. | No Combiner submitted. |
| Task 5: Hadoop Streaming Execution & Validation | Runs successfully using Hadoop Streaming; validates expected results and handles malformed input. | Runs successfully and includes reasonable validation. | Execution or validation is partial; some issues remain. | Execution fails or validation is inadequate. | No execution or validation evidence. |
| Task 6: Results Analysis & Documentation | Provides clear results, meaningful performance comparison, and well-organized report/code/commands. | Provides understandable results and documentation with most evidence. | Provides basic results and partial documentation or comparison. | Results are unclear and documentation/evidence is weak. | No report or deliverables submitted. |